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Inventor(s)	Fukumori; Takanori

Light-emitting element

Abstract

A light-emitting element includes, successively from a lower side to an upper side, a first n-side semiconductor layer, a first active layer, a first p-side semiconductor layer, a second n-side semiconductor layer, a second active layer, and a second p-side semiconductor layer, each made of a nitride semiconductor. The second n-side semiconductor layer contacts the first p-side semiconductor layer. The second n-side semiconductor layer includes, successively from a lower side to an upper side, a first layer including gallium, a second layer including aluminum and gallium, and a third layer including gallium and having a lower n-type impurity concentration than the first and second layers. A thickness of the first layer and a thickness of the second layer each is less than 50% of a thickness of the third layer.

Inventors:	Fukumori; Takanori (Tokushima, JP)
Applicant:	NICHIA CORPORATION (Anan, JP)
Family ID:	1000008820708
Assignee:	NICHIA CORPORATION (Anan, JP)
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Primary Examiner: Ojha; Ajay

Assistant Examiner: Kolahdouzan; Hajar

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims priority to Japanese Patent Application No. 2021-206192, filed on Dec. 20, 2021, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments relate to a light-emitting element.

BACKGROUND

(3) For example, JP-A 2004-128502 describes a light-emitting element including a first light-emitting part that includes a first n-type layer, a first active layer, and a first p-type layer, a tunnel junction layer located on the first light-emitting part, and a second light-emitting part that is located on the tunnel junction layer and includes a second n-type layer, a second active layer, and a second p-type layer.

SUMMARY

(4) According to one aspect of the present invention, a light-emitting element includes a first n-side semiconductor layer, a first active layer, a first p-side semiconductor layer, a second n-side semiconductor layer, a second active layer, and a second p-side semiconductor layer made of nitride semiconductors successively from a lower side to an upper side. The second n-side semiconductor layer contacts the first p-side semiconductor layer. The second n-side semiconductor layer includes, successively from a lower side to an upper side, a first layer including gallium, a second layer including aluminum and gallium, and a third layer including gallium and having a lower n-type impurity concentration than the first and second layers. A thickness of the first layer and a thickness of the second layer each is less than 50% of a thickness of the third layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view showing a light-emitting element **10** according to an embodiment;

(2) FIG. 2 is an enlarged cross-sectional view showing a second n-side semiconductor layer **121** of FIG. 1;

(3) FIG. 3 is an example of an energy band diagram of the light-emitting element **10** according to the embodiment;

(4) FIG. 4 is a flowchart showing the method for manufacturing the light-emitting element **10** according to the embodiment;

(5) FIG. 5 is a flowchart showing details of a step **S21** of forming the second n-side semiconductor layer **121**;

(6) FIG. 6 is a cross-sectional view illustrating manufacturing processes of the light-emitting

element **10** according to the embodiment;

(7) FIG. 7 is a cross-sectional view illustrating manufacturing processes of the light-emitting element **10** according to the embodiment;

(8) FIG. 8A is a cross-sectional view showing the layer configuration of a second n-side semiconductor layer **121A** of a light-emitting element **10A** according to examples 1 to 4;

(9) FIG. 8B is a cross-sectional view showing the layer configuration of a second n-side semiconductor layer **821** of a light-emitting element **80** according to a reference example 1;

(10) FIG. 8C is a cross-sectional view showing the layer configuration of a second n-side semiconductor layer **921** of a light-emitting element **90** according to a reference example 2;

(11) FIG. 9A is a graph showing a difference $\Delta V_{\text{sub.f}}$ between the forward voltage $V_{\text{sub.f}}$ and the forward voltage $V_{\text{sub.fref}}$ used as the reference for the light-emitting element **80** according to the reference example 1, the light-emitting element **90** according to the reference example 2, and the light-emitting element **10A** according to the examples; and

(12) FIG. 9B is a graph showing the values $P_{\text{sub.o}}/P_{\text{sub.oref}}$ of the normalized outputs $P_{\text{sub.o}}$ of the light-emitting element **80** according to the reference example 1, the light-emitting element **90** according to the reference example 2, and the light-emitting element **10A** according to the examples.

DETAILED DESCRIPTION

(13) Exemplary embodiments will now be described with reference to the drawings. The drawings are schematic or conceptual; and the relationships between the thickness and width of portions, the proportional coefficients of sizes among portions, etc., are not necessarily the same as the actual values thereof. Furthermore, the dimensions and proportional coefficients may be illustrated differently among drawings, even for identical portions. In the specification of the application and the drawings, components similar to those described in regard to a drawing therein above are marked with like reference numerals, and a detailed description is omitted as appropriate.

(14) For easier understanding of the following description, the arrangements and configurations of the portions are described using an XYZ orthogonal coordinate system. The X-axis, the Y-axis, and the Z-axis are orthogonal to each other. The direction in which the X-axis extends is taken as an “X-direction,” the direction in which the Y-axis extends is taken as a “Y-direction,” and the direction in which the Z-axis extends is taken as a “Z-direction.” Although up is taken as the Z-direction and down is taken as the opposite direction for easier understanding of the description, these directions are relative and are independent of the direction of gravity.

(15) FIG. 1 is a cross-sectional view showing a light-emitting element **10** according to the embodiment.

(16) FIG. 2 is an enlarged cross-sectional view showing a second n-side semiconductor layer **121** of FIG. 1.

(17) FIG. 3 is an example of an energy band diagram of the light-emitting element **10** according to the embodiment.

(18) As shown in FIG. 1, the light-emitting element **10** includes a substrate **11**, a semiconductor structure body **12**, an n-side electrode **13**, and a p-side electrode **14**.

(19) The substrate **11** has a flat plate shape. For example, the upper surface and the lower surface of the substrate **11** are substantially parallel to the X-Y plane. For example, the substrate **11** is made of sapphire ($\text{Al}_{0.2}\text{O}_{0.3}$). However, the substrate **11** may include another material, such as silicon (Si), silicon carbide (SiC), gallium nitride (GaN), etc. The semiconductor structure body **12** is located on the substrate **11**.

(20) The semiconductor structure body **12** is, for example, a stacked body in which multiple semiconductor layers made of nitride semiconductors are stacked. In this specification, a “nitride semiconductor” is a semiconductor that includes nitrogen, and typically includes all compositions of semiconductors of the chemical formula $\text{In}_x\text{Al}_y\text{Ga}_{1-x-y}\text{N}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, and $x+y \leq 1$) for which the composition ratios x and y are changed within the ranges respectively.

(21) The semiconductor structure body **12** includes a first light-emitting part **110** and a second light-emitting part **120** successively from a lower side to an upper side. The first light-emitting part **110** includes a first n-side semiconductor layer **112**, a first active layer **113**, and a first p-side semiconductor layer **114** successively from a lower side to an upper side. The first light-emitting part **110** further includes a foundation layer **111** located under the first n-side semiconductor layer **112**. The second light-emitting part **120** includes the second n-side semiconductor layer **121**, a second active layer **122**, and a second p-side semiconductor layer **123** successively from a lower side to an upper side. The components of the semiconductor structure body **12** will now be elaborated.

(22) The foundation layer **111** of the first light-emitting part **110** is located on the substrate **11**. The foundation layer **111** includes, for example, an undoped semiconductor layer. In the specification, “undoped” means that neither an n-type impurity nor a p-type impurity is intentionally doped. In other words, an undoped semiconductor layer is a semiconductor layer formed without supplying a raw material gas including an n-type impurity and/or a p-type impurity. The term “n-type impurity” means an impurity that forms donors. The term “p-type impurity” means an impurity that forms acceptors. There are cases in which an undoped semiconductor layer that is adjacent to a layer intentionally doped with an n-type impurity and/or a p-type impurity includes the n-type impurity and/or the p-type impurity due to diffusion into the undoped semiconductor layer from the adjacent layer, etc.

(23) The undoped semiconductor layer of the foundation layer **111** includes, for example, GaN. The first n-side semiconductor layer **112** is located on the foundation layer **111**. However, the first n-side semiconductor layer may be directly located on the substrate without including the foundation layer in the first light-emitting part.

(24) The first n-side semiconductor layer **112** includes one or more n-type semiconductor layers. The n-type semiconductor layer of the first n-side semiconductor layer **112** includes, for example, GaN doped with silicon (Si) as the n-type impurity. The n-type semiconductor layer of the first n-side semiconductor layer **112** may further include indium (In), aluminum (Al), etc. The n-type semiconductor layer of the first n-side semiconductor layer **112** may include germanium (Ge) as the n-type impurity.

(25) It is sufficient for the first n-side semiconductor layer **112** to have a function of supplying electrons to the first active layer **113**; and the first n-side semiconductor layer **112** may further include one or more undoped semiconductor layers. The undoped semiconductor layer of the first n-side semiconductor layer **112** includes, for example, GaN.

(26) The first n-side semiconductor layer **112** includes a first surface **112s1**, a second surface **112s2**, and a third surface **112s3**. The first surface **112s1** is substantially parallel to the X-Y plane. The second surface **112s2** is positioned higher than the first surface **112s1** and is substantially parallel to the X-Y plane. When viewed in top-view, the second surface **112s2** is next to the first surface **112s1** in the X-direction. The third surface **112s3** is positioned between the first surface **112s1** and the second surface **112s2** and is substantially parallel to the Y-Z plane. The first active layer **113** is located on the second surface **112s2**.

(27) The first active layer **113** has, for example, a multi-quantum well structure that includes multiple well layers and multiple barrier layers. The multiple well layers can include, for example, indium gallium nitride (InGaN). The multiple barrier layers can include, for example, GaN. The well layer and the barrier layer may be, for example, undoped semiconductor layers. At least portions of the well layer and the barrier layer may include an n-type impurity and/or a p-type impurity. The first p-side semiconductor layer **114** is located on the first active layer **113**.

(28) The first p-side semiconductor layer **114** includes, for example, one or more p-type semiconductor layers having a function of supplying holes to the first active layer **113**. The p-type semiconductor layer of the first p-side semiconductor layer **114** includes, for example, GaN doped with magnesium (Mg) as the p-type impurity. The p-type semiconductor layer of the first p-side

semiconductor layer **114** may further include Al.

(29) The first p-side semiconductor layer **114** may further include one or more undoped semiconductor layers. The undoped semiconductor layer of the first p-side semiconductor layer **114** includes, for example, GaN. The undoped semiconductor layer of the first p-side semiconductor layer **114** may further include Al. The first p-side semiconductor layer **114** may further include a p-type upper semiconductor layer that has a tunnel junction with the second n-side semiconductor layer **121**. The upper semiconductor layer includes, for example, GaN doped with Mg as the p-type impurity. For example, the p-type impurity concentration of the upper semiconductor layer is greater than the p-type impurity concentration of the p-type semiconductor layer positioned below the upper semiconductor layer in the first p-side semiconductor layer **114**. The second n-side semiconductor layer **121** is located on the first p-side semiconductor layer **114**.

(30) The second n-side semiconductor layer **121** contacts the first p-side semiconductor layer **114**. As shown in FIG. 2, the second n-side semiconductor layer **121** includes a first layer **121b**, a second layer **121c**, and a third layer **121d** successively from a lower side to an upper side.

According to the embodiment, the second n-side semiconductor layer **121** further includes a fourth layer **121a** located between the first layer **121b** and the first p-side semiconductor layer **114**, and a fifth layer **121e** located on the third layer **121d**.

(31) The fourth layer **121a** is located at the first p-side semiconductor layer **114**. The fourth layer **121a** is, for example, a GaN layer doped with Si as the n-type impurity. The fourth layer **121a** may further include In. In other words, the fourth layer **121a** may be an InGaN layer doped with Si as the n-type impurity. When the fourth layer **121a** is an InGaN layer, the value of the In composition ratio in the fourth layer **121a** is, for example, favorably not less than 10% and not more than 40%, and more favorably not less than 20% and not more than 40%.

(32) The first layer **121b** is located on the fourth layer **121a**. The first layer **121b** includes Ga. The first layer **121b** is, for example, a GaN layer doped with Si as the n-type impurity. The first layer **121b** may further include In. In other words, the first layer **121b** may be an InGaN layer doped with Si as the n-type impurity. When the first layer **121b** is an InGaN layer, it is favorable for the value of the In composition ratio in the first layer **121b** to be less than the value of the In composition ratio in the fourth layer **121a**. In such a case, it is favorable for the value of the In composition ratio in the first layer **121b** to be, for example, not less than 0.1% and not more than 3%.

(33) The second layer **121c** is located on the first layer **121b**. The second layer **121c** includes Al and Ga. According to the embodiment, the second layer **121c** is an AlGaN layer doped with Si as the n-type impurity. As shown in FIG. 3, the bandgap of the second layer **121c** is greater than the bandgap of the fourth layer **121a** and the bandgap of the first layer **121b**. The second layer **121c** may be, for example, an aluminum indium gallium nitride (AlInGaN) layer. The value of the Al composition ratio in the second layer **121c** is not less than 5% and not more than 35%.

(34) As shown in FIG. 2, the third layer **121d** is located on the second layer **121c**. The third layer **121d** includes Ga. The third layer **121d** is, for example, a GaN layer doped with Si as the n-type impurity.

(35) The fifth layer **121e** is located on the third layer **121d**. For example, the fifth layer **121e** has a function of supplying electrons to the second active layer **122**. The fifth layer **121e** can be, for example, a superlattice layer in which a GaN layer and an InGaN layer are alternately arranged. At least a portion of the semiconductor layer of the fifth layer **121e** may include an n-type impurity and/or a p-type impurity.

(36) The n-type impurity concentrations of the fourth layer **121a**, the first layer **121b**, the second layer **121c**, the third layer **121d**, and the fifth layer **121e** will now be described. Herein, the n-type impurity concentration of the fourth layer **121a** is also called the “fourth n-type impurity concentration”; the n-type impurity concentration of the first layer **121b** is also called the “first n-type impurity concentration”; the n-type impurity concentration of the second layer **121c** is also called the “second n-type impurity concentration”; the n-type impurity concentration of the third

layer **121d** is also called the “third n-type impurity concentration”; the n-type impurity concentration of the fifth layer **121e** is also called the “fifth n-type impurity concentration.”

(37) According to the embodiment, the magnitude relationship of the n-type impurity concentrations is fourth n-type impurity concentration>first n-type impurity concentration≥second n-type impurity concentration>third n-type impurity concentration. Also, second n-type impurity concentration>fifth n-type impurity concentration. However, the magnitude relationship of the n-type impurity concentrations is not limited to such a magnitude relationship. For example, the second n-type impurity concentration may be greater than the first n-type impurity concentration. Also, for example, the fifth n-type impurity concentration may be not less than the second n-type impurity concentration.

(38) The fourth n-type impurity concentration is, for example, not less than $2 \times 10^{20} \text{ cm}^{-3}$ and not more than $1 \times 10^{21} \text{ cm}^{-3}$. The first n-type impurity concentration is, for example, not less than $4 \times 10^{19} \text{ cm}^{-3}$ and not more than $2 \times 10^{20} \text{ cm}^{-3}$. The second n-type impurity concentration is, for example, not less than $2 \times 10^{19} \text{ cm}^{-3}$ and not more than $2 \times 10^{20} \text{ cm}^{-3}$. The third n-type impurity concentration is, for example, not less than $1 \times 10^{18} \text{ cm}^{-3}$ and not more than $2 \times 10^{19} \text{ cm}^{-3}$. The fifth n-type impurity concentration is, for example, not less than $1 \times 10^{18} \text{ cm}^{-3}$ and not more than $4 \times 10^{19} \text{ cm}^{-3}$.

(39) A thickness **t4** of the fourth layer **121a**, a thickness **t1** of the first layer **121b**, a thickness **t2** of the second layer **121c**, and a thickness **t3** of the third layer **121d** will now be described. Herein, “thickness” means the dimension of each layer in the vertical direction.

(40) According to the embodiment, the magnitude relationship of the thicknesses of the layers is thickness **t4**<thickness **t2**<thickness **t1**<thickness **t3**. However, the magnitude relationship of the thicknesses is not limited to such a magnitude relationship. For example, the thickness **t2** may be not less than the thickness **t1**.

(41) The thickness **t1** of the first layer **121b** and the thickness **t2** of the second layer **121c** each are less than 50% of the thickness **t3** of the third layer **121d**. The thickness **t2** of the second layer **121c** is not more than 40% of a total thickness **t12** of the first and second layers **121b** and **121c**. Here, total thickness **t12**=thickness **t1**+thickness **t2**. However, the thickness **t2** may be greater than 40% of the total thickness **t12**.

(42) The thickness **t4** of the fourth layer **121a** is, for example, not less than 0.1 nm and not more than 2 nm. The thickness **t1** of the first layer **121b** is, for example, not less than 10 nm and not more than 40 nm. The thickness **t2** of the second layer **121c** is, for example, not less than 10 nm and not more than 30 nm. The thickness **t3** of the third layer **121d** is, for example, not less than 80 nm and not more than 150 nm.

(43) The functions of the fourth layer **121a**, the first layer **121b**, the second layer **121c**, and the third layer **121d** will now be described.

(44) The fourth layer **121a** that has a high n-type impurity concentration is located at the vicinity of the p-n junction portion formed by the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121**. The carrier density at the vicinity of the p-n junction portion can be increased thereby, which reduces the width of the depletion layer at the p-n junction portion and makes electron tunneling easier.

(45) Because the fourth layer **121a** is an InGaN layer having a relatively high value of the In composition ratio, the bandgap of the fourth layer **121a** can be reduced. For example, as shown in FIG. 3, a portion of the conduction band has a low energy level at the vicinity of the p-n junction portion formed by the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121**. As a result, the tunnel effect easily occurs between the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121**. Although the details are described below, for example, the electrons **e** that exist in the valence band of the first p-side semiconductor layer **114** easily tunnel into the conduction band of the second n-side semiconductor layer **121** when the

forward voltage is applied to the light-emitting element **10**.

(46) There is a tendency for the crystallinity of a semiconductor layer to easily degrade when the value of the In composition ratio the semiconductor layer is relatively high or when the n-type impurity concentration is relatively high. According to the embodiment, the fourth layer **121a** has a higher n-type impurity and a larger value of the In composition ratio than the first layer **121b**, but the thickness **t4** of the fourth layer **121a** is less than the thickness of the first layer **121b**; therefore, the degradation of the crystallinity of the fourth layer **121a** can be reduced.

(47) On the other hand, by setting the first n-type impurity concentration of the first layer **121b** to be less than the fourth n-type impurity concentration and by setting the thickness **t1** of the first layer **121b** to be greater than the thickness **t4**, the carrier density at the vicinity of the p-n junction portion formed by the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121** can be increased while reducing the degradation of the crystallinity of the first layer **121b**. Because the In included in the first layer **121b** functions as a surfactant, the occurrence of point defects in the first layer **121b** can be reduced. The diffusion of the p-type impurity included in the first p-side semiconductor layer **114** into the semiconductor layers positioned higher than the first layer **121b**, such as the second layer **121c**, the third layer **121d**, the fifth layer **121e**, etc., can be reduced thereby.

(48) By forming the second layer **121c** that includes Al on the first layer **121b**, the point defects that occur when forming the semiconductor layers lower than the second layer **121c**, such as the first layer **121b**, the fourth layer **121a**, etc., are filled, and the surface state of the second layer **121c** can approach flat. By setting the second n-type impurity concentration of the second layer **121c** to be not more than the first n-type impurity concentration, the degradation of the crystallinity due to the second layer **121c** can be reduced while the first layer **121b** ensures the carrier density at the vicinity of the p-n junction portion formed by the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121**.

(49) Furthermore, the third layer **121d** that is located higher than the second layer **121c** has the thickness **t3** that is greater than the thickness **t1** and the third n-type impurity concentration that is less than the first n-type impurity concentration. The point defects that occur when forming the layers lower than the third layer **121d**, such as the first layer **121b**, the second layer **121c**, the fourth layer **121a**, etc., are further filled thereby, and the surface state of the third layer **121d** can further approach flat. The degradation of the crystallinity of the semiconductor layers positioned higher than the third layer **121d**, such as the fifth layer **121e**, the second active layer **122**, etc., can be reduced thereby. In particular, the luminous efficiency can be increased by reducing the degradation of the crystallinity of the second active layer **122**.

(50) However, the configuration of the second n-side semiconductor layer **121** is not limited to the configuration described above. For example, the second n-side semiconductor layer **121** may further include one or more undoped semiconductor layers. The undoped semiconductor layer of the second n-side semiconductor layer **121** includes, for example, GaN. Also, the second n-side semiconductor layer **121** may not include the fourth layer **121a**.

(51) The second active layer **122** has, for example, a multi-quantum well structure that includes multiple well layers and multiple barrier layers. The multiple well layers can include, for example, InGaN. The multiple barrier layers can include, for example, GaN. The well layer and the barrier layer may be, for example, undoped semiconductor layers. At least portions of the well layer and the barrier layer may include an n-type impurity and/or a p-type impurity. The second p-side semiconductor layer **123** is located on the second active layer **122**.

(52) The second p-side semiconductor layer **123** includes, for example, one or more p-type semiconductor layers. The p-type semiconductor layer of the second p-side semiconductor layer **123** includes, for example, GaN doped with Mg as the p-type impurity. The p-type semiconductor layer of the second p-side semiconductor layer **123** may further include Al.

(53) It is sufficient for the second p-side semiconductor layer **123** to have a function of supplying

holes to the second active layer **122**, and the second p-side semiconductor layer **123** may further include one or more undoped semiconductor layers. The undoped semiconductor layer of the second p-side semiconductor layer **123** includes, for example, GaN. The undoped semiconductor layer of the second p-side semiconductor layer **123** may further include Al.

(54) As shown in FIG. 1, the n-side electrode **13** is located on the first surface **112s1** of the first n-side semiconductor layer **112**. The n-side electrode **13** is electrically connected to the first n-side semiconductor layer **112**. The p-side electrode **14** is located on the second p-side semiconductor layer **123**. The p-side electrode **14** is electrically connected to the second p-side semiconductor layer **123**. The first active layer **113** and the second active layer **122** are caused to emit light by applying a forward voltage $V_{sub.f}$ between the n-side electrode **13** and the p-side electrode **14**.

(55) The light that is emitted by the first and second active layers **113** and **122** is, for example, ultraviolet light or visible light. The light emission peak wavelength of the first active layer **113** and the light emission peak wavelength of the second active layer **122** can be the same. For example, the first active layer **113** and the second active layer **122** may emit blue light. The light emission peak wavelength of the first active layer **113** and the light emission peak wavelength of the second active layer **122** may be different. For example, the first active layer **113** may emit blue light, and the second active layer **122** may emit green light. The light emission peak wavelength of the blue light is, for example, not less than 430 nm and not more than 490 nm. The light emission peak wavelength of the green light is, for example, not less than 500 nm and not more than 540 nm. The light emission peak wavelength of the ultraviolet light is not more than 400 nm.

(56) A reverse bias is applied between the second n-side semiconductor layer **121** and the first p-side semiconductor layer **114** when the forward voltage $V_{sub.f}$ is applied between the n-side electrode **13** and the p-side electrode **14**, that is, when a positive potential is applied to the p-side electrode **14** and a potential that is less than that of the p-side electrode **14** is applied to the n-side electrode **13**. Therefore, a tunnel effect due to the tunnel junction of the second n-side semiconductor layer **121** with the first p-side semiconductor layer **114** is utilized to cause a current to flow between the second n-side semiconductor layer **121** and the first p-side semiconductor layer **114**. Specifically, the current flows between the second n-side semiconductor layer **121** and the first p-side semiconductor layer **114** by electrons of the valence band of the first p-side semiconductor layer **114** tunneling to the conduction band of the second n-side semiconductor layer **121**. In other words, the second n-side semiconductor layer **121** has a tunnel junction with the first p-side semiconductor layer **114** if a current flows between the second n-side semiconductor layer **121** and the first p-side semiconductor layer **114** when the forward voltage $V_{sub.f}$ is applied to the light-emitting element **10**.

(57) By increasing the impurity concentrations of each conductivity type included in the p-type semiconductor layer and the n-type semiconductor layer forming the p-n junction of the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121**, the width of the depletion layer formed by the p-n junction can be reduced. When the voltage is applied, the electrons of the valence band of the first p-side semiconductor layer **114** tunnel into the conduction band of the second n-side semiconductor layer **121** more easily as the width of the depletion layer decreases.

(58) A method for manufacturing the light-emitting element **10** will now be described.

(59) FIG. 4 is a flowchart showing the method for manufacturing the light-emitting element **10** according to the embodiment.

(60) FIG. 5 is a flowchart showing details of a step **S21** of forming the second n-side semiconductor layer **121**.

(61) FIG. 6 is a cross-sectional view illustrating manufacturing processes of the light-emitting element **10** according to the embodiment.

(62) FIG. 7 is a cross-sectional view illustrating manufacturing processes of the light-emitting element **10** according to the embodiment.

(63) As shown in FIG. 4, the method for manufacturing the light-emitting element **10** includes a

step S1 of forming the first light-emitting part **110**, a step S2 of forming the second light-emitting part **120**, and a step S3 of forming the n-side electrode **13** and the p-side electrode **14**.

(64) For example, the first light-emitting part **110** and the second light-emitting part **120** that are included in the semiconductor structure body **12** are formed by MOCVD (metal organic chemical vapor deposition) in a furnace in which the pressure and the temperature can be adjusted. Specifically, the semiconductor structure body **12** is formed on the substrate **11** by supplying a carrier gas and a raw material gas to the furnace.

(65) For example, hydrogen (H.sub.2) gas, nitrogen (N.sub.2) gas, etc., can be used as the carrier gas.

(66) The raw material gas is appropriately selected according to the semiconductor layer to be formed. When a semiconductor layer that includes Ga is formed, for example, a raw material gas that includes Ga, such as trimethylgallium (TMG) gas, triethylgallium (TEG) gas, or the like is used. When forming a semiconductor layer that includes N, for example, a raw material gas that includes N, such as ammonia (NH.sub.3) gas or the like is used. When forming a semiconductor layer that includes Al, for example, a raw material gas that includes Al, such as trimethylaluminum (TMA) gas or the like is used. When forming a semiconductor layer that includes In, for example, a raw material gas that includes In, such as trimethylindium (TMI) or the like is used. When forming a semiconductor layer that includes Si, for example, a gas that includes Si, such as monosilane (SiH.sub.4) gas or the like is used. When forming a semiconductor layer that includes Mg, for example, a raw material gas that includes Mg, such as bis cyclopentadienyl magnesium (Cp.sub.2Mg) gas or the like is used. Hereinbelow, the supply of a raw material gas that includes one element and a raw material gas that includes another element to the furnace also is called simply “supplying a raw material gas that includes one element and another element.” The processes will now be elaborated.

(67) First, the step S1 of forming the first light-emitting part **110** is performed.

(68) The step S1 of forming the first light-emitting part **110** includes a step S11 of forming the foundation layer **111**, a step S12 of forming the first n-side semiconductor layer **112**, a step S13 of forming the first active layer **113**, and a step S14 of forming the first p-side semiconductor layer **114**.

(69) In the step S11 of forming the foundation layer **111**, a carrier gas and a raw material gas that correspond to the foundation layer **111** are supplied to the furnace. Thereby, the foundation layer **111** is formed on the substrate **11**.

(70) In the step S12 of forming the first n-side semiconductor layer **112**, a carrier gas and a raw material gas that correspond to the first n-side semiconductor layer **112** are supplied to the furnace. Thereby, the first n-side semiconductor layer **112** is formed on the foundation layer **111**.

(71) The step S13 of forming the first active layer **113** supplies a carrier gas and a raw material gas corresponding to the first active layer **113** to the furnace. Thereby, the first active layer **113** is formed on the first n-side semiconductor layer **112**.

(72) In the step S14 of forming the first p-side semiconductor layer **114**, a carrier gas and a raw material gas that correspond to the first p-side semiconductor layer **114** are supplied to the furnace. Thereby, the first p-side semiconductor layer **114** is formed on the first active layer **113**.

(73) Thus, as shown in FIG. 6, the first light-emitting part **110** that includes the foundation layer **111**, the first n-side semiconductor layer **112**, the first active layer **113**, and the first p-side semiconductor layer **114** is formed on the substrate **11**.

(74) Then, the step S2 of forming the second light-emitting part **120** is performed.

(75) As shown in FIG. 4, the step S2 of forming the second light-emitting part **120** includes the step S21 of forming the second n-side semiconductor layer **121**, a step S22 of forming the second active layer **122**, and a step S23 of forming the second p-side semiconductor layer **123**.

(76) According to the embodiment as shown in FIG. 5, the step S21 of forming the second n-side semiconductor layer **121** includes a step S21a of forming the fourth layer **121a**, a step S21b of

forming the first layer **121b**, a step **S21c** of forming the second layer **121c**, a step **S21d** of forming the third layer **121d**, and a step **S21e** of forming the fifth layer **121e**.

(77) In the step **S21a** of forming the fourth layer **121a**, a carrier gas, a raw material gas that includes In, Ga, and N, and a raw material gas that includes Si as an n-type impurity are supplied. Thereby, the fourth layer **121a** that is made of an InGaN layer doped with Si is formed on the first p-side semiconductor layer **114**.

(78) In the step **S21b** of forming the first layer **121b**, a carrier gas, a raw material gas that includes Ga and N, and a raw material gas that includes Si as an n-type impurity are supplied. Thereby, the first layer **121b** that is made of a GaN layer doped with Si is formed on the fourth layer **121a**. At this time, the flow rates of the raw material gas including Ga, the raw material gas including N, and the raw material gas including Si are adjusted so that the first n-type impurity concentration is less than the fourth n-type impurity concentration.

(79) Also, the thickness **t1** of the first layer **121b** is formed to be less than 50% of the thickness **t3** of the third layer **121d**. At this time, a raw material gas that includes In also may be supplied. For example, the thicknesses of the semiconductor layers can be adjusted by appropriately modifying the formation times of the semiconductor layers.

(80) In the step **S21c** of forming the second layer **121c**, a carrier gas, a raw material gas that includes Al, Ga, and N, and a raw material gas that includes Si as an n-type impurity are supplied. Thereby, the second layer **121c** that is made of an AlGaN layer doped with Si is formed on the first layer **121b**. At this time, the flow rates of the raw material gas including Al, the raw material gas including Ga, the raw material gas including N, and the raw material gas including Si are adjusted so that the second n-type impurity concentration is not more than the first n-type impurity concentration. However, as described above, the second n-type impurity concentration may be greater than the first n-type impurity concentration. Also, the thickness **t2** of the second layer **121c** is formed to be less than 50% of the thickness **t3** of the third layer **121d**.

(81) In the step **S21d** of forming the third layer **121d**, a carrier gas, a raw material gas that includes Ga and N, and a raw material gas that includes Si as an n-type impurity are supplied. Thereby, the third layer **121d** that is made of a GaN layer doped with Si is formed on the second layer **121c**. At this time, the flow rates of the raw material gas including Ga, the raw material gas including N, and the raw material gas including Si are adjusted so that the third n-type impurity concentration is less than the second n-type impurity concentration. Also, the thickness **t3** of the third layer **121d** is formed to be greater than the thickness **t1** of the first layer **121b**.

(82) In the step **S21e** of forming the fifth layer **121e**, a GaN layer that is doped with Si as an n-type impurity and an InGaN layer that is doped with Si as an n-type impurity are alternately formed on the third layer **121d**. The GaN layer that is doped with Si is formed by supplying a carrier gas, a raw material gas that includes Ga and N, and a raw material gas that includes Si to the furnace. The InGaN layer that is doped with Si is formed by supplying a carrier gas, a raw material gas that includes In, Ga, and N, and a raw material gas that includes Si to the furnace.

(83) The step **S22** of forming the second active layer **122** supplies a carrier gas and a raw material gas that correspond to the second active layer **122** to the furnace. Thereby, the second active layer **122** is formed on the second n-side semiconductor layer **121**.

(84) In the step **S23** of forming the second p-side semiconductor layer **123**, a carrier gas and a raw material gas that correspond to the second p-side semiconductor layer **123** are supplied to the furnace. Thereby, the second p-side semiconductor layer **123** is formed on the second active layer **122**.

(85) Thus, as shown in FIG. 7, the second light-emitting part **120** that includes the second n-side semiconductor layer **121**, the second active layer **122**, and the second p-side semiconductor layer **123** is formed on the first light-emitting part **110**.

(86) Then, the step **S3** of forming the n-side electrode **13** and the p-side electrode **14** is performed.

(87) In the step **S3** of forming the n-side electrode **13** and the p-side electrode **14**, first, the first and

third surfaces **112s1** and **112s3** of the first n-side semiconductor layer **112** are exposed at the first active layer **113**, the first p-side semiconductor layer **114**, and the second light-emitting part **120** by removing a portion of the semiconductor structure body **12** as shown in FIG. **1**. For example, the portion of the semiconductor structure body **12** can be removed by selectively etching by using a resist.

(88) Then, the n-side electrode **13** is formed on the exposed first surface **112s1**. The p-side electrode **14** is formed on the second p-side semiconductor layer **123**. For example, the n-side electrode **13** and the p-side electrode **14** can be formed by sputtering or vapor deposition.

(89) Thus, the light-emitting element **10** can be obtained. However, the method for manufacturing the light-emitting element **10** is not limited to the method described above. For example, the method for manufacturing the light-emitting element **10** may not include the step **S11** of forming the foundation layer **111**, and the first n-side semiconductor layer **112** may be directly formed on the substrate **11**.

(90) Effects of the embodiment will now be described.

(91) The light-emitting element **10** according to the embodiment includes the first n-side semiconductor layer **112**, the first active layer **113**, the first p-side semiconductor layer **114**, the second n-side semiconductor layer **121** contacting the first p-side semiconductor layer **114**, the second active layer **122**, and the second p-side semiconductor layer **123** that are made of nitride semiconductors successively from a lower side to an upper side. The second n-side semiconductor layer **121** includes, successively from a lower side to an upper side, the first layer **121b** that includes Ga, the second layer **121c** that includes Al and Ga, and the third layer **121d** that includes Ga and has a lower n-type impurity concentration than the first and second layers **121b** and **121c**. The thickness **t1** of the first layer **121b** and the thickness **t2** of the second layer **121c** each are less than 50% of the thickness **t3** of the third layer **121d**.

(92) According to the light-emitting element **10** described above, the carrier concentration of the vicinity of the p-n junction portion formed by the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121** can be increased by the first layer **121b** having a higher n-type impurity concentration than the third layer **121d**. The second layer **121c** that includes Al fills the point defects occurring when forming the semiconductor layers lower than the second layer **121c**, such as the first layer **121b**, etc., and the surface state of the second layer **121c** can approach flat. The point defects are further filled by the third layer **121d** that has a lower n-type impurity concentration than the first and second layers **121b** and **121c**, and the surface state of the third layer **121d** can approach flat even more. The degradation of the crystallinity can be reduced by setting the thicknesses of the first and second layers **121b** and **121c** that have higher n-type impurity concentrations than the third layer **121d** to be less than the thickness **t3** of the third layer **121d**. Thus, the degradation of the crystallinity of the second active layer **122** formed on the second n-side semiconductor layer **121** can be reduced. The luminous efficiency of the light-emitting element **10** can be increased thereby.

(93) The thickness **t2** of the second layer **121c** is less than the thickness **t1** of the first layer **121b**. Therefore, the crystallinity can be improved by the second layer **121c** while reducing the degradation of the forward voltage $V_{sub.f}$ due to an increase of the thickness **t2** of the second layer **121c**.

(94) The thickness **t2** of the second layer **121c** is not more than 40% of the total thickness **t12** of the first and second layers **121b** and **121c**. Therefore, the crystallinity can be improved by the second layer **121c** while reducing the degradation of the crystallinity of the first layer **121b**.

(95) The n-type impurity concentration of the second layer **121c** is not more than the n-type impurity concentration of the first layer **121b**. Therefore, the crystallinity can be improved by the second layer **121c** while the first layer **121b** ensures the carrier density at the vicinity of the p-n junction portion formed by the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121**.

(96) The first layer **121b** may further include In. In such a case, the surface state of the first layer **121b** can approach flat because the In included in the first layer **121b** functions as a surfactant.

(97) The second n-side semiconductor layer **121** further includes the fourth layer **121a** located between the first layer **121b** and the first p-side semiconductor layer **114**. The fourth layer **121a** includes In and Ga and has a higher n-type impurity concentration than the first and second layers **121b** and **121c**. According to such a fourth layer **121a**, the carrier density at the vicinity of the p-n junction portion formed by the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121** can be increased. Because the bandgap of the fourth layer **121a** can be reduced, the tunnel effect easily occurs between the first p-side semiconductor layer **114** and the second n-side semiconductor layer **121**.

EXAMPLES

(98) Examples and reference examples will now be described.

(99) FIG. **8A** is a cross-sectional view showing the layer configuration of a second n-side semiconductor layer **121A** of a light-emitting element **10A** according to the examples 1 to 4.

(100) FIG. **8B** is a cross-sectional view showing the layer configuration of a second n-side semiconductor layer **821** of a light-emitting element **80** according to a reference example 1.

(101) FIG. **8C** is a cross-sectional view showing the layer configuration of a second n-side semiconductor layer **921** of a light-emitting element **90** according to a reference example 2.

(102) FIG. **8A** shows the layer configuration, and the actual magnitude relationship of the thicknesses of the layers may be different.

(103) The light-emitting element **10A** according to the examples 1 to 4 differed from the light-emitting element **10** according to the embodiment described above in that the second n-side semiconductor layer **121A** was made of the first layer **121b**, the second layer **121c**, the third layer **121d**, and the fifth layer **121e** and did not include the fourth layer **121a**; otherwise, the layer configuration was similar to that of the light-emitting element **10** according to the embodiment. The light-emitting element **10A** according to the examples 1 to 4 was made to have the thickness **t1** of the first layer **121b** and the thickness **t2** of the second layer **121c** shown in the following table.

(104) TABLE-US-00001

Example	Example 1	Example 2	Reference example 1	Reference example 2
First layer (Lowermost layer)	15 nm	17.5 nm	20 nm	25 nm
Second layer	35 nm	35 nm	15 nm	10 nm
Third layer	100 nm	100 nm	100 nm	100 nm

(105) The light-emitting element **80** according to the reference example 1 differed from the light-emitting element **10A** according to the examples 1 to 4 in that the first layer **121b** and the second layer **121c** were replaced with a GaN layer doped with Si (hereinbelow, called a “lowermost layer **821a**”); otherwise, the layer configuration was made to be common with the light-emitting element **10A** according to the examples 1 to 4.

(106) The light-emitting element **90** according to the reference example 2 differed from the light-emitting element **10A** according to the examples 1 to 4 in that the first layer **121b** and the second layer **121c** were replaced with an AlGaIn layer doped with Si (hereinbelow, called a “lowermost layer **921a**”); otherwise, the layer configuration was made to be common with the light-emitting element **10A** according to the examples 1 to 4. The method for manufacturing the light-emitting elements **10A**, **80**, and **90** will now be elaborated.

(107) First, the foundation layer **111** that included an undoped GaN layer and had a thickness of about 5 μm was formed on the substrate **11** made of sapphire.

(108) Then, the first n-side semiconductor layer **112** that had a thickness of about 5.5 μm and included an undoped GaN layer and a GaN layer doped with Si was formed on the foundation layer **111**.

(109) Continuing, the first active layer **113** that had a thickness of about 50 nm and included multiple barrier layers including undoped GaN layers and multiple well layers including undoped InGaIn layers was formed on the first n-side semiconductor layer **112**. The first active layer **113**

was formed to include seven pairs of the barrier layer and the well layer.

(110) Then, the first p-side semiconductor layer **114** that included a GaN layer doped with Mg and had a thickness of about 15 nm was formed on the first active layer **113**.

(111) Continuing, on the first p-side semiconductor layer **114**, the second n-side semiconductor layer **821** was formed in the reference example 1, the second n-side semiconductor layer **921** was formed in the reference example 2, and the second n-side semiconductor layer **121A** was formed in the examples 1 to 4.

(112) The second n-side semiconductor layer **821** of the reference example 1 included the lowermost layer **821a**, the third layer **121d**, and the fifth layer **121e** successively from a lower side to an upper side. The lowermost layer **821a** was made of a GaN layer doped with Si, and a thickness **t81** of the lowermost layer **821a** was about 35 nm. The third layer **121d** was made of a GaN layer doped with Si, and the thickness **t3** of the third layer **121d** was about 100 nm. The third n-type impurity concentration of the third layer **121d** was less than the n-type impurity concentration of the lowermost layer **821a**. The n-type impurity concentration of the lowermost layer **821a** was equal to the n-type impurity concentration of the first layer **121b**. The fifth layer **121e** included a stacked body in which a GaN layer doped with Si and an InGaN layer doped with Si were alternately stacked, and had a thickness of about 56 nm.

(113) The second n-side semiconductor layer **921** of the reference example 2 included the lowermost layer **921a**, the third layer **121d**, and the fifth layer **121e** successively from a lower side to an upper side. The lowermost layer **921a** was made of an AlGaN layer doped with Si, and a thickness **t91** of the lowermost layer **921a** was about 35 nm. The value of the Al composition ratio of the lowermost layer **921a** was about 13%. The n-type impurity concentration of the lowermost layer **921a** was substantially equal to the n-type impurity concentration of the lowermost layer **821a**. The n-type impurity concentration of the lowermost layer **921a** was equal to the n-type impurity concentration of the second layer **121c**. The third layer **121d** of the reference example 2 was formed similarly to the third layer **121d** of the reference example 1, and the fifth layer **121e** of the reference example 2 was formed similarly to the fifth layer **121e** of the reference example 1.

(114) The second n-side semiconductor layer **121A** of the examples 1 to 4 included the first layer **121b**, the second layer **121c**, the third layer **121d**, and the fifth layer **121e** successively from a lower side to an upper side.

(115) The first layer **121b** was made of a GaN layer doped with Si. The thickness **t1** of the first layer **121b** in the example 1 was 15 nm, the thickness **t1** of the first layer **121b** in the example 2 was 17.5 nm, the thickness **t1** of the first layer **121b** in the example 3 was 20 nm, and the thickness **t1** of the first layer **121b** in the example 4 was 25 nm.

(116) The second layer **121c** was made of an AlGaN layer doped with Si. The value of the Al composition ratio of the second layer **121c** was about 13%. The second n-type impurity concentration of the second layer **121c** was less than the first n-type impurity concentration of the first layer **121b**. The thickness **t2** of the second layer **121c** in the example 1 was 20 nm, the thickness **t2** of the second layer **121c** in the example 2 was 17.5 nm, the thickness **t2** of the second layer **121c** in the example 3 was 15 nm, and the thickness **t2** of the second layer **121c** in the example 4 was 10 nm. The third layer **121d** of the examples 1 to 4 was formed similarly to the third layer **121d** of the reference example 1, and the fifth layer **121e** of the examples 1 to 4 was formed similarly to the fifth layer **121e** of the reference example 1.

(117) Then, the second active layer **122** that had a thickness of about 50 nm and included multiple barrier layers including undoped GaN layers and multiple well layers including undoped InGaN layers was formed on the second n-side semiconductor layers **121A**, **821**, and **921**. The second active layer **122** was formed to include seven pairs of the barrier layer and the well layer.

(118) Continuing, the second p-side semiconductor layer **123** that included a GaN layer doped with Mg and had a thickness of about 11 nm was formed on the second active layer **122**.

(119) Then, portions of the first n-side semiconductor layer **112**, the first active layer **113**, the first

p-side semiconductor layer **114**, the second p-side semiconductor layers **121A**, **821**, and **921**, the second active layer **122**, and the second p-side semiconductor layer **123** were removed, the n-side electrode **13** was formed on the exposed first n-side semiconductor layer **112**, and the p-side electrode **14** was formed on the second p-side semiconductor layer.

(120) FIG. **9A** is a graph showing a difference $\Delta V_{\text{sub.f}}$ between the forward voltage $V_{\text{sub.f}}$ and the forward voltage $V_{\text{sub.fref}}$ used as the reference for the light-emitting element **80** according to the reference example 1, the light-emitting element **90** according to the reference example 2, and the light-emitting element **10A** according to the examples.

(121) FIG. **9B** is a graph showing the values $P_{\text{sub.o}}/P_{\text{sub.oref}}$ of the normalized outputs $P_{\text{sub.o}}$ of the light-emitting element **80** according to the reference example 1, the light-emitting element **90** according to the reference example 2, and the light-emitting element **10A** according to the examples.

(122) The forward voltages $V_{\text{sub.f}}$ were measured for the light-emitting element **80** according to the reference example 1, the light-emitting element **90** according to the reference example 2, and the light-emitting element **10A** according to the examples 1 to 4 that were made. The results are shown in FIG. **9A**. The forward voltage $V_{\text{sub.f}}$ of the reference example 1 was used as the reference forward voltage $V_{\text{sub.fref}}$, and the vertical axis of FIG. **9A** is the values of the reference forward voltage $V_{\text{sub.fref}}$ subtracted from the measured forward voltages $V_{\text{sub.f}}$. In other words, $\Delta V_{\text{sub.f}} = V_{\text{sub.f}} - V_{\text{sub.fref}}$.

(123) The outputs $P_{\text{sub.o}}$ were measured for the light-emitting element **80** according to the reference example 1, the light-emitting element **90** according to the reference example 2, and the light-emitting element **10A** according to the examples 1 to 4 that were made. The results are shown in FIG. **9B**. The output $P_{\text{sub.o}}$ of the reference example 1 was used as the reference output $P_{\text{sub.oref}}$, and the vertical axis of FIG. **9B** is the values of the measured outputs $P_{\text{sub.o}}$ normalized by dividing by the reference output $P_{\text{sub.oref}}$.

(124) The light-emitting element **90** according to the reference example 2 differed from the light-emitting element **80** according to the reference example 1 in that the lowermost layer **921a** was an AlGaIn layer instead of a GaN layer. Although the output $P_{\text{sub.o}}$ of the light-emitting element **90** according to the reference example 2 was greater than the output $P_{\text{sub.o}}$ of the light-emitting element **80** according to the reference example 1 as shown in FIG. **9B**, the forward voltage $V_{\text{sub.f}}$ of the light-emitting element **90** according to the reference example 2 was greater than the forward voltage $V_{\text{sub.f}}$ of the light-emitting element **80** according to the reference example 1 as shown in FIG. **9A**. Accordingly, it was found that when simply replacing the GaN layer with an AlGaIn layer, the output $P_{\text{sub.o}}$ can be increased, but the forward voltage $V_{\text{sub.f}}$ is undesirably increased.

(125) The light-emitting element **10A** according to the examples 1 to 4 differed from the light-emitting element **90** according to the reference example 2 in that the first layer **121b** made of the GaN layer and the second layer **121c** that was made of an AlGaIn layer and located on the first layer **121b** were included instead of the lowermost layer **921a** made of an AlGaIn layer. According to the examples 1 to 4, the thickness t_1 of the first layer **121b** and the thickness t_2 of the second layer **121c** each were less than 50% of the thickness of the third layer **121d**. As shown in FIG. **9B**, the outputs $P_{\text{sub.o}}$ of the light-emitting element **10A** according to the examples 1 to 4 were less than the output $P_{\text{sub.o}}$ of the light-emitting element **90** according to the reference example 2 but greater than the output $P_{\text{sub.o}}$ of the light-emitting element **80** according to the reference example 1. As shown in FIG. **9A**, the forward voltages $V_{\text{sub.f}}$ of the light-emitting element **10A** according to the examples 1 to 4 were less than the forward voltage $V_{\text{sub.f}}$ of the light-emitting element **90** according to the reference example 2 but equal to the forward voltage $V_{\text{sub.f}}$ of the light-emitting element **80** according to the reference example 1. In other words, the high forward voltage $V_{\text{sub.f}}$ was reduced to a value about equal to the forward voltage $V_{\text{sub.f}}$ of the reference example 1 by replacing the lower part of the lowermost layer **921a** with a GaN layer (the first layer **121b**) and by setting the thickness t_1 of the first layer **121b** and the thickness t_2 of the second layer **121c** each to

be less than 50% of the thickness of the third layer **121d**. Accordingly, it is favorable for the second n-side semiconductor layer **121** to include both the first and second layers **121b** and **121c** and for the thickness **t1** of the first layer **121b** and the thickness **t2** of the second layer **121c** each to be less than 50% of the thickness of the third layer **121d**.

(126) In particular, among the light-emitting elements **10A** according to the examples 1 to 4, the forward voltage $V_{sub.f}$ was a minimum and the output $P_{sub.o}$ was a maximum in the light-emitting element **10A** according to the example 4. The thickness **t2** of the second layer **121c** was less than the thickness **t1** of the first layer **121b** in the light-emitting element **10A** according to the example 4. More specifically, in the light-emitting element **10A** according to the example 4, the thickness **t2** (10 nm) of the second layer **121c** was not more than 40% of the total thickness **t12** (35 nm) of the first and second layers **121b** and **121c**. Accordingly, it is favorable for the thickness **t2** of the second layer **121c** to be less than the thickness **t1** of the first layer **121b**. It is favorable for the thickness **t2** of the second layer **121c** to be not more than 40% of the total thickness **t12** of the first and second layers **121b** and **121c**.

(127) According to embodiments described above, a light-emitting element that has high luminous efficiency can be provided.

Claims

1. A light-emitting element, comprising: successively from a lower side to an upper side, a first n-side semiconductor layer, a first active layer, a first p-side semiconductor layer, a second n-side semiconductor layer, a second active layer, and a second p-side semiconductor layer, each of which is made of a nitride semiconductor; wherein: the second n-side semiconductor layer contacts the first p-side semiconductor layer; the second n-side semiconductor layer comprises, successively from a lower side to an upper side: a first layer comprising gallium and having a first n-type impurity concentration, a second layer comprising aluminum and gallium and having a second n-type impurity concentration, and a third layer comprising gallium and having a third n-type impurity concentration lower than the first n-type impurity concentration and the second n-type impurity concentration; and a thickness of the first layer and a thickness of the second layer are each less than 50% of a thickness of the third layer.
2. The element according to claim 1, wherein: the thickness of the second layer is less than the thickness of the first layer.
3. The element according to claim 1, wherein: the thickness of the second layer is not more than 40% of a total thickness of the first and second layers.
4. The element according to claim 2, wherein: the thickness of the second layer is not more than 40% of a total thickness of the first and second layers.
5. The element according to claim 1, wherein: the second n-type impurity concentration is not more than the first n-type impurity concentration.
6. The element according to claim 2, wherein: the second n-type impurity concentration is not more than the first n-type impurity concentration.
7. The element according to claim 3, wherein: the second n-type impurity concentration is not more than the first n-type impurity concentration.
8. The element according to claim 1, wherein: a value of a composition ratio of aluminum of the second layer is not less than 5% and not more than 35%.
9. The element according to claim 2, wherein: a value of a composition ratio of aluminum of the second layer is not less than 5% and not more than 35%.
10. The element according to claim 3, wherein: a value of a composition ratio of aluminum of the second layer is not less than 5% and not more than 35%.
11. The element according to claim 1, wherein: the first layer further comprises indium.
12. The element according to claim 2, wherein: the first layer further comprises indium.

13. The element according to claim 1, wherein: the second n-side semiconductor layer further comprises a fourth layer located between the first layer and the first p-side semiconductor layer; and the fourth layer comprises indium and gallium and has a fourth n-type impurity concentration higher than the first n-type impurity concentration and the second n-type impurity concentration.
 14. The element according to claim 2, wherein: the second n-side semiconductor layer further comprises a fourth layer located between the first layer and the first p-side semiconductor layer; and the fourth layer comprises indium and gallium and has a fourth n-type impurity concentration higher than the first n-type impurity concentration and the second n-type impurity concentration.
 15. The element according to claim 1, wherein: the thickness of the first layer is not less than 10 nm and not more than 40 nm.
 16. The element according to claim 2, wherein: the thickness of the first layer is not less than 10 nm and not more than 40 nm.
 17. The element according to claim 1, wherein: the thickness of the second layer is not less than 10 nm and not more than 30 nm.
 18. The element according to claim 15, wherein: the thickness of the second layer is not less than 10 nm and not more than 30 nm.
 19. The element according to claim 1, wherein: the thickness of the third layer is not less than 80 nm and not more than 150 nm.
 20. The element according to claim 1, wherein: the thickness of the third layer is not less than 80 nm and not more than 150 nm.
 21. The element according to claim 18, wherein: the thickness of the third layer is not less than 80 nm and not more than 150 nm.
 22. The element according to claim 1, wherein: the first layer and the third layer are gallium nitride layers doped with an n-type impurity; and the second layer is an aluminum gallium nitride layer doped with an n-type impurity.
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